From a patient's devices to a forecast on their phone and a ranked list for the care team.

System overview

Data sources

Continuous glucose monitor
A reading every 5 minutes. Demo sensor, or a Dexcom Clarity / LibreView CSV export.

Smartwatch or band
Steps and walks, heart rate, sleep.

Health records (ABHA)
HbA1c and medicines. Lab report or prescription upload.

Meal log
Typed, spoken or photographed meals.

Blood-pressure cuff
Home readings.

Consent, source by source
Each stream is switched on separately. The twin and the care team see only what the patient shares.

ingest

Server · Node.js + Fastify

one process; fitting in worker threads

API · REST + Server-Sent Events
Sign-in with cookie sessions (scrypt passwords) · consent checks on every read · patient, care-team, AI and system routes · live updates pushed to the browser over SSE.

Twin service (hub)
- Forecast cache per patient
- Alerts, snack plans, care-team escalation
- Triage: rank by predicted 6-hour risk
- What-if simulations, timeline, messages

Fit worker pool
Worker threads run the twin engine for each person:
- personal fit (MAP, Nelder–Mead)
- unlogged-meal detection
- self-calibration on 10 evenings

Re-fits nightly at 03:00.

agent tools

Live loop + demo clock
Simulated time at 1×, 15× or 60×. Advances the synthetic patients and writes a new CGM reading every 5 simulated minutes. Forecasts and alerts update on each reading.

AI service
Meal parsing (text, voice, photo) · lab-report extraction · plain-language explanations · SBAR handoff notes · drafted messages · "Ask your twin" agent with 5 tools that call the twin service. Offline fallbacks: Indian food table and templates.

SQLite (better-sqlite3)
Users and sessions · consents and sources · events (meals, walks, sleep, doses) · CGM readings · fitted twins and track records · alerts · messages · chat history · demo clock.

HTTPS JSON

+ SSE live

Browser · React + Vite

Patient
- Live twin: forecast, alert, reasons
- Log a meal, Ask your twin
- What-if simulator
- Messages
- Connect data, Build my twin

Care team
- Patients ranked by predicted risk
- Handoff note, drafted messages
- Any patient's twin

Hand-built SVG charts · English, हिन्दी, ಕನ್ನಡ · 360–1440 px

Google Gemini API
gemini-3.8-flash (configurable) via @google/genai. JSON-schema output, vision for photos and PDFs, function calling. Optional: the app runs without it.

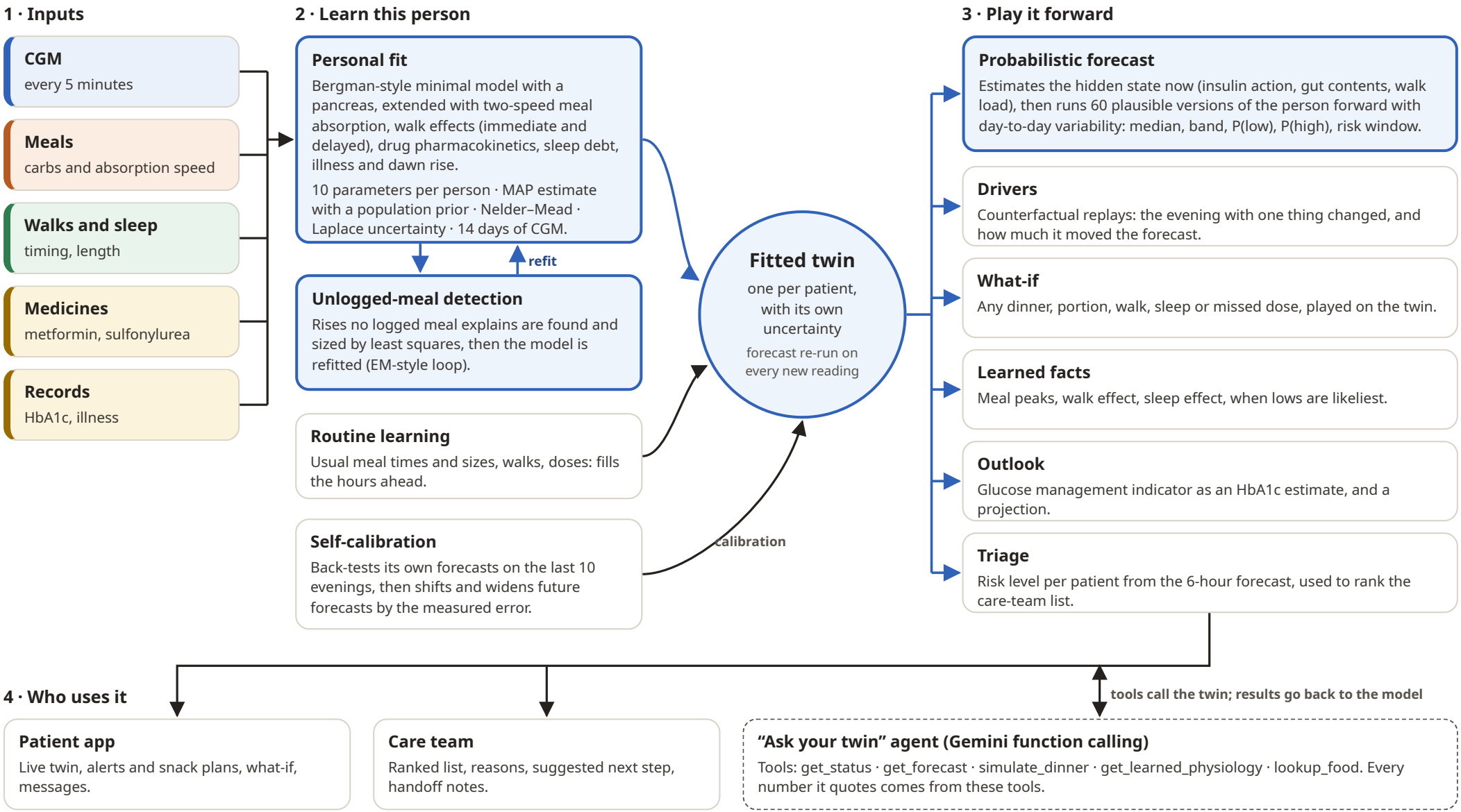
optional · falls back offline

request / data flow · twin engine · optional external service

Research prototype · synthetic data · not a medical device

 Plain TypeScript in **shared/**, used by the server, the browser and the tests.

The twin engine



Research prototype · synthetic data · not a medical device